MIL-H-6536C(ASG)
22 AUGUST 1968

Superseding
MIL-H-6536B(ASG)
12 October 1956

MILITARY SPECIFICATION

HORN AND VALVE, CREW MEMBERS' RELIEF, PRESSURIZED AIRCRAFT

This specification has been approved by the Department of the Air Force and by the Naval Air Systems Command.

1. SCOPE

1.1 Scope. - This specification covers two types of relief horn and valve units.

1.2 Classification. - Relief horn and valve units shall be one of the following types, as specified (see 6.2):

Type I - Straight
Type II - 90-degree, right-angle bend

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-B-626	Brass, Leaded and Nonleaded, Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-W-423	Wire, Steel, Corrosion-Resisting
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Setup

Military

MIL-P-116	Preservation, Methods of

FSC 1680

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831	Test Reports; Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction inspection. - This specification makes provision for preproduction inspection (see 4.3).

3.2 Selection of specifications and standards. - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. -

3.3.1 Brass. - The brass used shall conform to QQ-B-626.

3.3.2 Nickel plating. - The nickel plating shall be in accordance with QQ-N-290, class 2.

3.3.3 Spring steel. - The spring used for the release trigger shall be constructed of stainless spring steel, conforming to QQ-W-423, condition C.

3.3.4 Cadmium plating. - Cadmium plating shall conform to QQ-P-416, type II, class B, or equivalent.

3.3.5 Plastic. - The plastic shall be equal to or interchangeable with lumarith X-T-13507MH, color No. 0-2165-A, Celanese Celluloid Corporation, New York, or Tenite II-201A961MS, Tennessee Eastman Corporation, Kingsport, Tennessee.

3.4 Design and construction. - The horn and valve assembly shall consist of a plastic cone-type horn with a spring-tensed valve. The unit shall be so constructed that it can be held and operated by one hand. The valve shall open when the lever is depressed, and shall automatically close when the lever is released. The valve shaft housing shall be large enough to prevent a buildup of negative pressure in the horn chamber when a vacuum is applied across the outlet with the horn chamber plugged.

3.5 Performance. -

3.5.1 Valve leakage. - When tested as specified in 4.6.2, the air leakage past the valve seat shall not exceed 0.20 cubic feet per minute (cfm).

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3.5.2 Vacuum. - With the horn chamber plugged and the valve open, it shall not be possible to induce an absolute pressure of less than 13.7 psia in the horn in 3 minutes' time with the horn outlet connected to a 12.0 psia source.

3.5.3 Valve actuating force. - A force not to exceed 1 pound shall be required to open the valve with the lever when a differential pressure of not less than 12 psia is applied to the closed valve (see 4.6.4).

3.5.4 High and low temperature. - When tested as specified in 4.6.5, the horn and valve assemblies shall show no evidence of cracking, chipping, or warping.

3.5.5 Fracture or breakage. - The horn and valve assembly shall not break or fracture upon being dropped from a height of not less than 4 feet to a flat metal surface while being maintained at a temperature of 0° ±5° F (see 4.6.6).

3.5.6 Drainage. - When tested as specified in 4.6.7, the time required to drain 50 milliliters of water shall be not more than 3 seconds.

3.6 Dimensions. - The type I valve and horn assembly shall conform to the dimensions specified on figure 1. The type II valve and horn assembly shall conform to the dimensions specified on figure 2.

3.7 Weight. - The weight of the horn and valve assembly shall not exceed 0.20 pound.

3.8 Finish. - All brass shall be nickel plated and all noncorrosion-resistant steel shall be cadmium plated, or equivalent.

3.9 Identification of product. - Horn and valve assemblies shall be marked for identification in accordance with MIL-STD-130 (marking shall not be indent stamped).

3.10 Workmanship. - Horn and valve assemblies shall be constructed in accordance with good commercial practice and shall be free from any defect that might affect appearance or functionability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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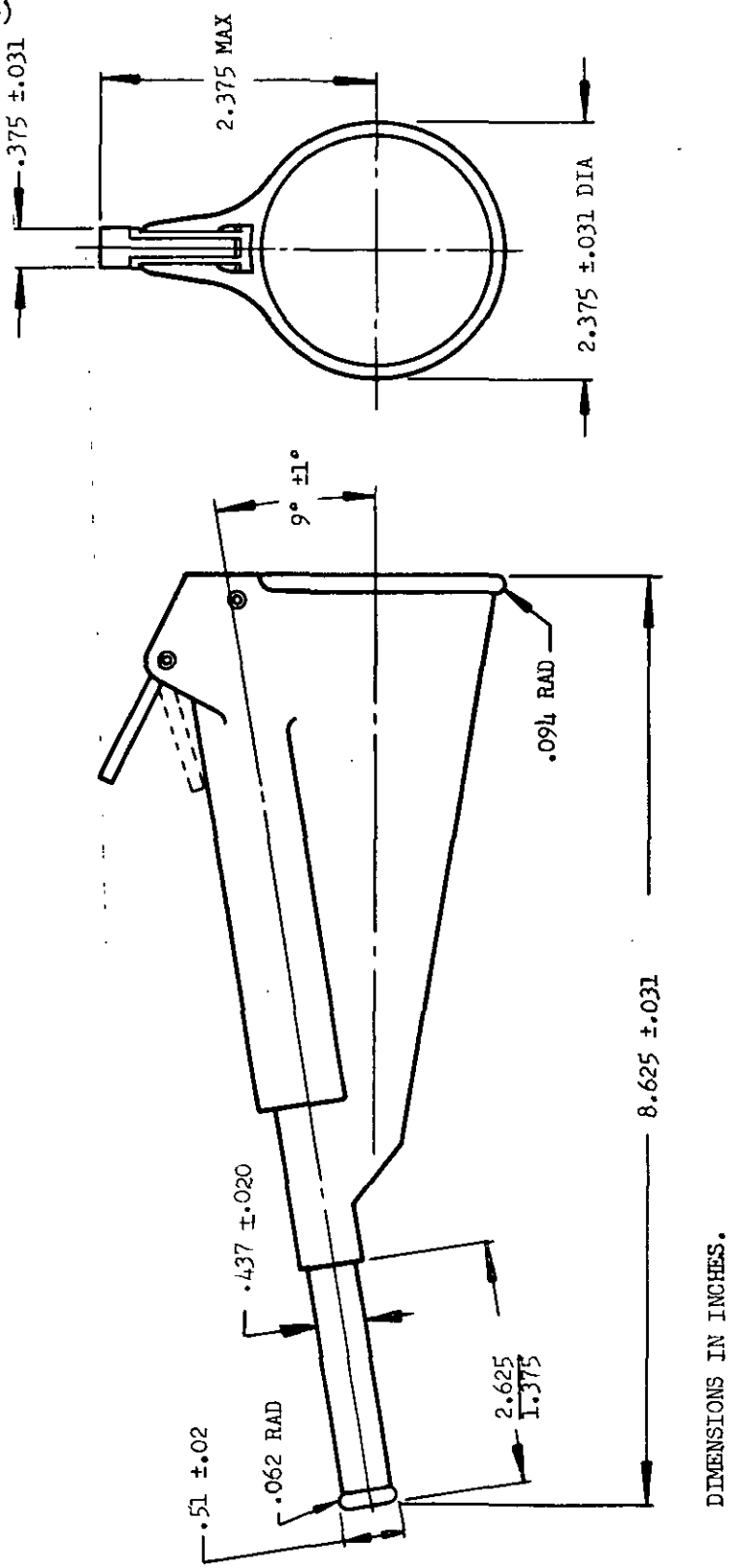
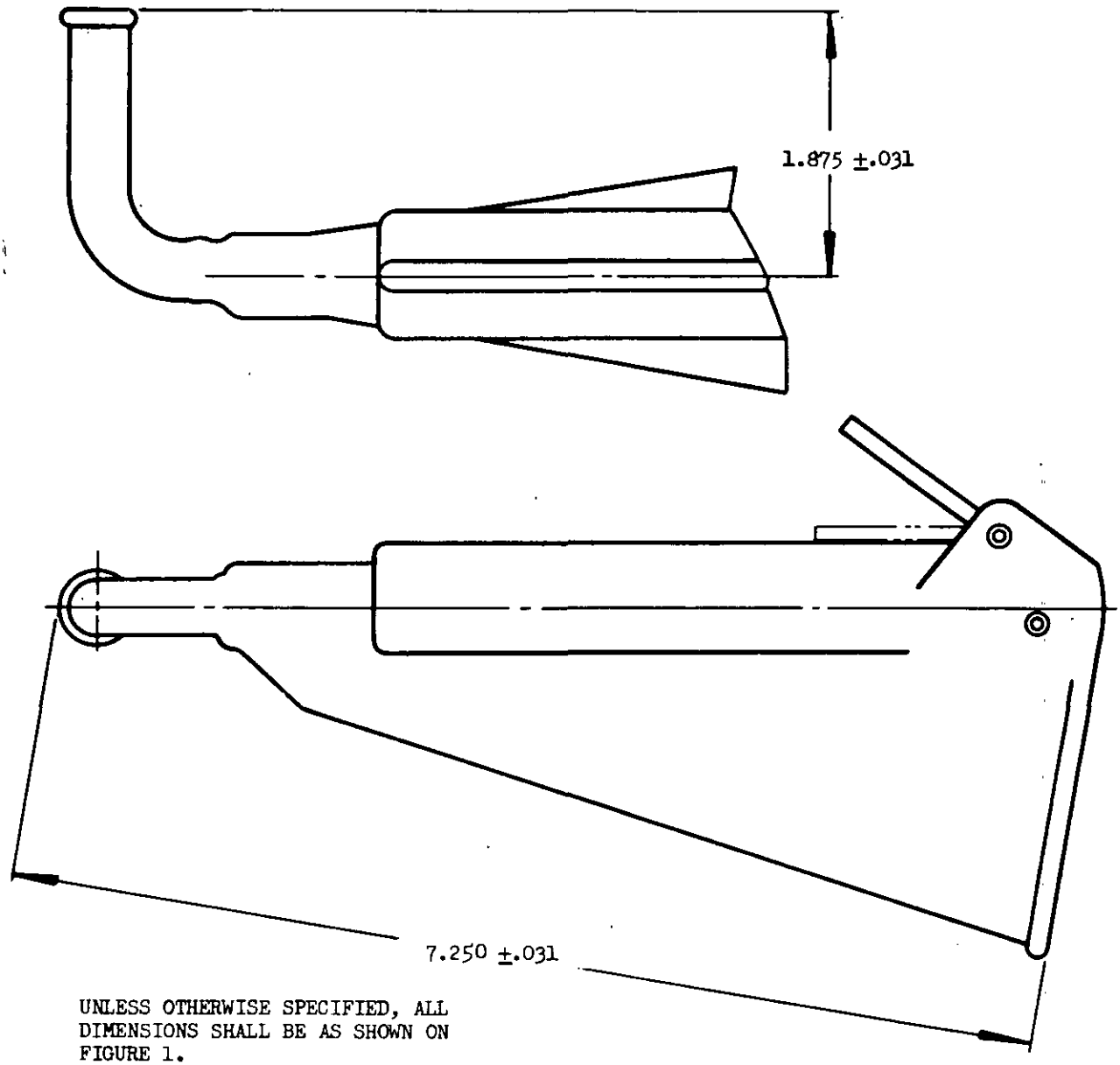


FIGURE 1. Type I horn and valve assembly, fliers' relief tube

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UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS SHALL BE AS SHOWN ON FIGURE 1.

DIMENSIONS IN INCHES.

FIGURE 2. Type II horn and valve assembly, fliers' relief tube, 90-degree right bend

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4.2 Classification of inspections. - The examination and testing of horn and valve assemblies shall be classified as follows:

(a)	Preproduction inspection	(4.3)
(b)	Quality conformance inspection	(4.4)

4.3 Preproduction inspection. -

4.3.1 Sampling instructions. - The preproduction samples shall consist of two horn and valve assemblies representative of the production of horn and valve assemblies, and shall be tested by the contractor.

4.3.2 Tests. - Preproduction inspection shall consist of the examination and tests specified in 4.6.

4.3.3 Preproduction test report. - After the contractor completes the preproduction inspection, he shall prepare a test report in accordance with MIL-STD-831.

4.3.4 Preproduction test sample for the procuring activity. - Along with the preproduction test report, the contractor shall submit one sample to the procuring activity.

4.4 Quality conformance inspection. - The quality conformance inspection shall consist of individual tests and sampling tests (see 4.4.1 and 4.4.2).

4.4.1 Individual tests. - Each horn and valve assembly shall be subjected to the examination of product specified in 4.6.1.

4.4.2 Sampling tests. - Two horn and valve units shall be selected at random from each lot of 500, or fraction thereof on the order, and subjected to the following tests:

(a)	Valve leakage	(4.6.2)
(b)	Vacuum	(4.6.3)
(c)	Valve actuating force	(4.6.4)
(d)	High and low temperature	(4.6.5)
(e)	Fracture or breakage	(4.6.6)
(f)	Drainage	(4.6.7)

4.4.3 Rejection and retest. - When one or more items from a lot fail to meet this specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.4.3.1 Individual tests may continue. - For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until and unless it is determined that the lot meets all the requirements of this specification.

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4.5 Test conditions. - Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77° ±18° F and a relative humidity of 80 percent or less. Where tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance shall be made for the change in instrument reading.

4.6 Inspection methods. -

4.6.1 Examination of product. - Each horn and valve assembly shall be examined to determine conformance with the dimensions, workmanship and construction requirements.

4.6.2 Valve leakage. - The air leakage past the valve seat shall not exceed 0.20 cfm when the closed valve is subjected to a differential pressure of not less than 12 psia for not less than 3 minutes.

4.6.3 Vacuum. - The performance requirement specified in 3.5.2 is applicable for inspection.

4.6.4 Valve actuating force. - The force required to open the valve shall not exceed 1 pound when a differential pressure of not less than 12 psia is applied to the closed valve.

4.6.5 High and low temperature. - One sample horn and valve assembly shall be subjected to a temperature of +160° ±5° F for not less than 4 hours. The second sample horn and valve assembly shall be subjected to a temperature of -65° ±5° F, for not less than 4 hours. Any evidence of cracking, chipping, or warping shall be cause for rejection.

4.6.6 Fracture or breakage. - The sample horn and valve units shall be dropped from a height of not less than 4 feet to a flat metal surface while the assembly is being maintained at a temperature of 0° ±5° F. Evidence of fracture or breakage shall be cause for rejection.

4.6.7 Drainage. - With the horn in a vertical position and the valve fully closed, the horn shall be filled with 50 milliliters of water. Upon opening the valve, the 50 milliliters shall drain from the horn in not more than 3 seconds.

4.7 Preparation for delivery. - Preservation, packaging, packing, and marking shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. - Horn and valve units shall be preserved and packaged in accordance with MIL-P-116, Method III, one each in a unit container conforming to PPP-B-676.

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5.1.2 Level C. - Level C shall afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice, provided the latter meets the requirements of this level.

5.2 Packing. - Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. - Horn and valve units preserved and packaged as specified in 5.1.1 shall be packed in overseas type shipping containers conforming to PPP-B-601 or weather-resistant type shipping containers conforming to PPP-B-636. As far as practicable, shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the specification. Containers shall be closed and strapped in accordance with the specification and appendix thereto.

5.2.2 Level B. - Horn and valve units preserved and packed as specified in 5.1.1 shall be packed in domestic type shipping containers conforming to PPP-B-601 or PPP-B-636. Shipping containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, shipping containers shall be of uniform shape and size and contain identical quantities. The gross weight of each shipping container shall not exceed the weight limitation of the specification. Containers shall be closed and strapped in accordance with the specification and appendix thereto.

5.2.3 Level C. - Level C shall afford adequate protection at the lowest rate against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Marking of shipments. - Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - The horn and valve assemblies covered by this specification are intended for use in pressurized aircraft for relief of crew members. They are designed for use with Bracket - Flyer's Relief Tube, Vertical (MS22025), or Bracket - Flyer's Relief Tube, Horizontal (AN8020, No. 2).

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Preproduction requirements (see 4.3.3 and 4.3.4).
(d) Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

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6.3 Marginal indicia. - The margins of this specification have not been marked, due to the extensiveness of changes.

Custodians:
Navy - AS
Air Force - 82

Reviewer activity:
Air Force - 82

Preparing activity:
Air Force - 82

Project No. 1680-0050

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-H-6536C(ASG) Horn and Valve, Crew Members' Relief, Pressurized AC

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

( ) Direct Government Contract ( ) Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE?

   ( ) YES ( ) NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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